

MASS-APP-014-V03 - Reasoning, Insight & Decision Intelligence

Document Control

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1. Purpose

V03 defines the Decision Intelligence room: the governed environment in which MASS organizes evidence, synthesizes information, detects patterns, compares scenarios, explains reasoning, surfaces risks and opportunities, and presents practical options to authorized decision owners.

V03 improves human decisions. It does not make autonomous decisions, authorize execution, or replace executive accountability.

2. Scope

V03 includes decision cases, questions, evidence chains, multi-source synthesis, insights, recommendations, decision scoring, opportunity and risk signals, scenario comparison, forecasts, organizational patterns, historical outcome comparison, explainable reasoning traces, human decisions, NOVA and POPS advisory requests, cross-department intelligence, plugin extension contracts, and governed agent collaboration.

It excludes autonomous approval, autonomous execution, hidden chain-of-thought storage, source-system mutation, cross-tenant learning, enterprise analytics ownership, knowledge ownership, workflow ownership, and unrestricted agent behavior.

3. Platform Consumption Map

Dependency	Consumed capability	Delegated responsibility
APP-014-V01	Intelligence workspaces, sessions, prompt composition	Creative intelligence foundation
APP-014-V02	Organizational memory, citations, confidence, history	Memory stewardship
APP-013	Read-only projects and design artifacts	Design artifact stewardship
ENG-003/004/005	Identity, authorization, events	Platform controls
ENG-007	Knowledge retrieval	Enterprise knowledge authority
ENG-009	AI and governed agent orchestration	Models, prompts, tools, guardrails
ENG-017/019/020/022/023	Domain intelligence	Domain authority
ENG-024	Analytics, patterns, forecasts, benchmarks	Enterprise analytics authority
ENG-027	Executive information and graph references	Executive intelligence authority

Owned responsibilities are decision-case structure, evidence chains, case synthesis, explainable insights, recommendation records, advisory scoring, scenario presentation, human decision capture, and extension or agent contribution records.

Delegated responsibilities are source facts to owning applications, memory to V02 and ENG-007, analytics to ENG-024, AI and agents to ENG-009, execution to ENG-006 and owning capabilities, and authorization to ENG-004.

Published events: decision.case.created, decision.evidence.added, decision.insight.generated, decision.risk.detected, decision.opportunity.detected, decision.recommendation.ready, decision.human.recorded, decision.outcome.recorded.

Consumed events: organizational.memory.preserved, organizational.memory.source.stale, enterprise.analytics.updated, enterprise.risk.updated, enterprise.opportunity.updated, financial.forecast.updated, operational.state.changed.

V03 Role Mapping

V03 roles extend the APP-014 role hierarchy; they do not replace it.

V03 role	APP-014 baseline	Relationship and authority
Decision Owner	Steward, which extends V01 Editor	Specializes Steward for assigned decision cases. May control eligible case lifecycle, freeze review packages, close cases, and request handoffs. Cannot record the final executive decision unless separately assigned Executive Approver.
Executive Approver	Administrator, which maps to V01 Admin	Specializes Administrator for a defined decision scope. May record HumanDecision and DecisionAmendment records when ENG-004 confirms case-specific approval authority. It does not grant tenant-wide administration unless that baseline role is separately assigned.

Viewer, Contributor, Steward, and Administrator retain their V02 definitions. Case assignment narrows access; it never broadens tenant or source authorization.

4. Architecture and Lifecycle

Authorized sources enter through typed gateways. Evidence is normalized into case-scoped references while retaining source ownership. Synthesis and analytic requests route to ENG-024 or ENG-009. The Reasoning Service assembles a structured, explainable trace. The Recommendation Service presents options. A human decision owner alone records the selected outcome.

Lifecycle: Draft -> Evidence Gathering -> Analysis -> Ready for Review -> Decided -> Outcome Pending -> Closed -> Archived.

- Draft defines owner, question, scope, deadline, and affected domains.
- Evidence Gathering collects authorized source references and citations.
- Analysis creates synthesis, signals, scenarios, forecasts, scores, and traces.
- Ready for Review freezes the reviewed recommendation package.
- Decided records an authorized human selection, rationale, conditions, and dissent.
- Outcome Pending monitors authorized outcome references without executing work.
- Closed compares expected and verified outcomes and preserves learning through V02.
- Archived retains the complete institutional record.

Reopening a decided case creates a new decision cycle linked to the previous cycle.

5. Capability Architecture

Capability	Purpose	Inputs	Outputs	Dependencies	V1 implementation	Future evolution
Executive Reasoning	Frame a decision in owner language	Question, constraints, evidence, objectives	Trace, uncertainty, options	V01, V02, ENG-009, ENG-027	Structured analysis with human review	Longitudinal executive reasoning

Capability	Purpose	Inputs	Outputs	Dependencies	V1 implementation	Future evolution
Evidence Chains	Prove claims against sources	Memories, domain records, citations	Ordered supporting and contradicting evidence	V02, ENG-007	Manual and assisted assembly	Automated gap mapping
Multi-source Synthesis	Combine domain intelligence	Authorized evidence	Agreements, conflicts, omissions	ENG-007, ENG-009, ENG-024	Governed synthesis request	Continuous federated synthesis
Recommendations	Present practical options	Synthesis, scenarios, constraints	Ranked candidates	ENG-009, ENG-024	Advisory options only	Adaptive option generation
Decision Scoring	Expose evaluation criteria	Options, weights, confidence	Scores and uncertainty	V02, ENG-024	Transparent weighted scoring	Calibrated decision-class models
Opportunity Detection	Surface beneficial possibilities	Growth, relationship, operations signals	Opportunity candidates	ENG-017, ENG-019, ENG-024	Rule and analytics signals	Ecosystem opportunity composition
Risk Detection	Surface exposure and downside	Risk, finance, operations, compliance	Risk candidates and mitigations	ENG-022, ENG-024, risk owners	Rule and analytics signals	Dynamic risk propagation
Scenario Comparison	Compare plausible options	Assumptions, variables, constraints	Scenario outcomes	ENG-024	User-defined comparisons	Simulation-backed scenarios
Forecast Framework	Present expected ranges	Metrics, drivers, horizon	Forecast snapshots	ENG-024	Consume approved forecasts	Probabilistic and causal forecasts
Pattern Recognition	Find recurring behavior	Historical cases and outcomes	Evidence-backed pattern candidates	V02, ENG-024	Tenant-local comparison	Longitudinal pattern libraries
Historical Learning	Compare decisions and outcomes	Closed cases, verified outcomes	Lessons and precedent references	V02, ENG-007	Human-verified precedent capture	Outcome-calibrated guidance
Explainability	Explain why options exist	Evidence, scoring, rules	Structured reasoning trace	ENG-009, ENG-011	Claims, citations, assumptions	Expanded explanation audits
Human Approval	Preserve accountable authority	Reviewed package and principal	Human decision record	ENG-003, ENG-004	Mandatory human decision	Remains mandatory
NOVA Advisory	Add enterprise-intelligence perspective	Authorized case package	Advisory analysis	ENG-009, ENG-027	Explicit user request	Broader cross-domain advisory
POPS Advisory	Add stewardship perspective	Authorized case package	Governance and continuity advisory	ENG-009, ENG-027	Explicit user request	Expanded institutional stewardship
Plugin Extensions	Admit specialized analysis	Redacted request	Evidence, signal, score, or option candidate	Future Plugin Library, ENG-004	Contract and registry shape	Approved plugin ecosystem
Agent Collaboration	Coordinate bounded specialists	Task, evidence, agent role	Signed contribution	ENG-009	Sequential user-invoked calls	Governed parallel collaboration

Future evolution is architectural placement, not V1 authorization.

6. Evidence Chains and Synthesis

An EvidenceChain is case-scoped and revisioned. Every link identifies a claim, source reference, source revision, citation locator, relationship, confidence, and display order. Relationships are supports, contradicts, qualifies, contextualizes, or unknown.

Every recommendation claim resolves to evidence or is labeled assumption. Stale sources remain visible and reduce confidence. Evidence retains its constitutional owner.

Synthesis identifies sources included and excluded, agreement, contradictions, missing evidence, recency, confidence limitations, assumptions, affected departments, and unresolved questions. It is a case artifact. Institutional preservation is delegated to V02.

7. Recommendations and Scoring

A Recommendation is an advisory option with expected benefit, cost, risks, dependencies, reversibility, time horizon, and evidence chain.

DecisionScore dimensions are evidence quality, expected impact, urgency, strategic alignment, reversibility, execution feasibility, risk exposure, and uncertainty. Each records value, weight, rationale, and evidence. The aggregate is reproducible and cannot conceal missing evidence.

Rank order never constitutes approval. An authorized human may select any option, reject all, request evidence, or defer.

8. Risk, Opportunity, Scenario, and Forecast Intelligence

RiskSignal and OpportunitySignal are candidates, not authoritative domain records. Each contains origin, affected scope, severity or value, likelihood, time sensitivity, evidence, confidence, and owner routing. Confirmed signals are handed to their constitutional owner.

A Scenario defines assumptions, variables, constraints, horizon, and option. ScenarioOutcome records expected benefits, costs, risks, dependencies, and uncertainty ranges.

A ForecastReference points to an ENG-024 output and records method, baseline date, horizon, low/base/high range, assumptions, drivers, confidence, and source version. V03 does not independently calculate enterprise forecasts.

V1 presents user-defined scenarios and consumed analytics. Simulation remains future evolution under an approved analytical capability.

9. Patterns and Historical Learning

PatternCandidate identifies repeated conditions, actions, and outcomes across authorized tenant-local history. It includes sample size, time range, confidence, counterexamples, and source cases.

Closing a case records verified outcomes. Lessons are preserved through V02 only after human confirmation. V03 never trains across tenants and never treats correlation as causation.

10. Explainable AI

V03 stores a structured ReasoningTrace, not hidden model chain-of-thought. It contains the question, claims, evidence links, assumptions, scoring criteria, alternatives, uncertainties, model interaction references, human edits, and final rationale.

Users receive concise explanations tied to citations. ENG-009 retains provider interaction governance. Sensitive internal model reasoning is neither requested nor persisted.

11. Human Authority

Only an authenticated and authorized Decision Owner or Executive Approver may record a HumanDecision. The selected option, rationale, conditions, dissent, timestamp, and authorization reference become immutable. Corrections create a DecisionAmendment.

No recommendation, score, NOVA response, POPS response, plugin contribution, or agent contribution may transition a case to Decided.

12. NOVA and POPS

NOVA and POPS are advisory request types executed through ENG-009, not downstream systems.

NOVA evaluates enterprise context, dependencies, patterns, and consequences. POPS evaluates stewardship, constitutional alignment, continuity, and responsibility boundaries. Both receive only authorized context and return signed advisory evidence. Neither approves or executes.

13. Plugin and Agent Extension Points

DecisionExtension permits evidence_candidate, signal_candidate, score_candidate, and option_candidate outputs. Each invocation records extension identifier, version, tenant, case, permissions, redacted input hash, output, citations, confidence, duration, and error.

Extensions are read-only toward decision records and sources. The host validates accepted contributions. V1 provides contract and registry shape only; no marketplace or unrestricted runtime is authorized.

AgentTask identifies one specialist role, bounded objective, allowed evidence, output schema, timeout, and human requester. ENG-009 executes it. V1 supports sequential user-invoked contributions. Future parallel agents remain governed by human approval and ENG-009.

14. Security

- Every entity is tenant-bound and protected by concrete RLS.
- Child entities use composite tenant foreign keys.
- Polymorphic references require tenant and authorization validation.
- Reviewed packages and HumanDecision records are database-immutable.
- Case access cannot exceed the most restrictive cited source.
- Plugin and agent inputs are minimized and redacted.
- Cross-tenant learning and case comparison are prohibited.

15. Data Model

Entity	Purpose	Core integrity
DecisionCase	Stable decision container	Tenant lifecycle and V01 workspace
DecisionQuestion	Revised decision frame	One active revision
EvidenceItem	Authorized source reference	Source tenant and lifecycle validation
EvidenceChain	Revised claim network	Immutable when reviewed
EvidenceChainLink	Claim-to-evidence link	Direct evidence and chain FKs
Synthesis	Multi-source interpretation	Sources, omissions, conflicts preserved
Insight	Explainable observation	Evidence-backed or assumed
Recommendation	Advisory option	Cannot approve or execute
DecisionScore	Reproducible score	Values, weights, rationale, evidence
RiskSignal	Candidate risk	Domain-owner handoff
OpportunitySignal	Candidate opportunity	Domain-owner handoff
Scenario	Assumptions and constraints	Case and option scoped
ScenarioOutcome	Expected outcome range	Direct scenario reference
ForecastReference	Consumed ENG-024 forecast	Immutable source version
PatternCandidate	Historical pattern	Sample, confidence, counterexamples
ReasoningTrace	Structured explanation	No hidden chain-of-thought
HumanDecision	Authorized decision	Database-immutable
DecisionAmendment	Correction without overwrite	Original decision FK
ExtensionInvocation	Plugin contribution	Idempotent and read-only
AgentContribution	Specialist output	ENG-009 reference

16. API Contracts

Method	Route	Purpose
GET, POST	/decision-cases	List or create cases
GET, PATCH	/decision-cases/{id}	Retrieve or edit case
POST	/decision-cases/{id}/archive	Archive case
GET, POST	/decision-cases/{id}/questions	List or revise question
GET, POST	/decision-cases/{id}/evidence	List or add evidence
DELETE	/decision-cases/{id}/evidence/{evidenceId}	Remove eligible evidence
GET, POST	/decision-cases/{id}/evidence-chains	List or create chains
POST	/decision-cases/{id}/syntheses	Request synthesis
GET, POST	/decision-cases/{id}/insights	List or record insights
GET, POST	/decision-cases/{id}/recommendations	List or create options
POST	/recommendations/{id}/scores	Record score
GET	/decision-cases/{id}/risk-signals	List risks
GET	/decision-cases/{id}/opportunity-signals	List opportunities
POST	/decision-cases/{id}/signals/detect	Request signal detection
GET, POST	/decision-cases/{id}/scenarios	List or create scenarios
POST	/decision-cases/{id}/scenarios/compare	Compare scenarios
GET, POST	/decision-cases/{id}/forecasts	List or attach forecasts
GET	/decision-cases/{id}/patterns	List patterns
POST	/decision-cases/{id}/patterns/detect	Request pattern analysis
GET	/decision-cases/{id}/reasoning-traces	List explanations
POST	/decision-cases/{id}/nova-advisory	Request NOVA
POST	/decision-cases/{id}/pops-advisory	Request POPS
POST	/decision-cases/{id}/agent-tasks	Request bounded agent
POST	/decision-cases/{id}/extensions/{extensionId}	Invoke extension
POST	/decision-cases/{id}/ready-for-review	Freeze package
POST	/decision-cases/{id}/decisions	Record human decision
POST	/decisions/{id}/amendments	Record amendment
POST	/decision-cases/{id}/outcomes	Attach outcome
POST	/decision-cases/{id}/close	Close case

Mutating requests require idempotency keys. Advisory, extension, and agent failures leave case state unchanged.

17. Folder Structure and Migrations

V03 adds decision-cases, evidence, synthesis, insights, recommendations, scoring, signals, scenarios, forecasts, patterns, reasoning, decisions, extensions, and agents. Gateways alone call V02, ENG-009, ENG-024, NOVA, POPS, or plugin contracts.

Migrations continue from 029 through 041: decision cases, questions, evidence, chains, synthesis, insights, recommendations, scores, signals, scenarios, forecasts, patterns, reasoning, decisions, extensions, agents, RLS, and enforcement triggers.

The implementation-grade PostgreSQL contract is MASS-APP-014-V03_Migration_Reference.sql. It defines all V03 tables, columns, foreign keys, indexes, constraints, triggers, and RLS policies and shall be split into migrations 029 through 041 without weakening those controls.

18. Failure Behavior

- Inaccessible evidence: reject attachment.
- Stale evidence: retain with warning and reduce confidence.

- Analytics unavailable: record failure; never fabricate.
- AI unavailable: fail the request only.
- Extension or agent timeout: record failure; preserve case.
- Contradictory evidence: surface conflict.
- Missing human authority: reject decision transition.
- Unverified outcome: retain pending; do not learn from it.

19. V1 and Future Boundary

V1 implements human-created cases, typed evidence, governed synthesis, transparent scoring, user-defined scenarios, consumed forecasts, signal candidates, structured explanations, sequential advisories, human decisions, and verified outcomes.

Future evolution may add richer simulations, calibrated models, continuous pattern monitoring, approved plugins, and parallel specialist agents. These capabilities have architectural homes but are not V1 runtime requirements.

20. Acceptance Verification

- Every requested capability has a defined architectural home and V1/future boundary.
- V03 consumes rather than duplicates V01, V02, ENG-009, ENG-024, and ENG-027.
- Recommendations and scores remain advisory.
- Human authority is mandatory and immutable.
- Explainability uses structured evidence, not hidden chain-of-thought.
- Plugins and agents cannot mutate decisions or sources.
- Cross-tenant intelligence is prohibited.

21. Constitutional Boundary Statement

V03 organizes evidence and presents decision intelligence. It does not own the facts, relationships, analytics, risks, opportunities, finances, operations, communications, or knowledge it consumes. It does not authorize or execute decisions. Authority remains with constitutionally accountable humans and owning enterprise capabilities.